AN INTERSECTION OF IDEALS IS COMPUTED BY AN H1-REGULAR SEQUENCE MODULO AN IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Koszul complex of a sequence over a ring, and of the same sequence over a quotient. used at proof lines 7481-7485 building the short exact sequence 0 to I tensor K to K to K of A/I to 0. Provides the three complexes and the horizontal maps whose exactness drives the argument.
- H_1-regularity of a sequence means vanishing of the first Koszul homology. used at proof lines 7486-7487, where the hypothesis that the images are H_1-regular makes H_1 of the right hand complex zero. Is the single hypothesis that turns the long exact sequence into the injectivity statement the proof needs.
- The kernel of the first Koszul differential is the ideal generated by the sequence. used at proof lines 7488-7493 to translate injectivity of the cokernel map into the identity I cap (g_1, ..., g_m) = I(g_1, ..., g_m). Performs the final identification of the homological statement with the assertion about ideals.

Source reference. Original lemma 0665: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7924–7924. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let A be a ring. Let $I \subset A$ be an ideal. Let $g_1, \ldots, g_m$ be a sequence in A whose image in A/I is H_1-regular. Then $I \cap (g_1, \ldots, g_m) = I(g_1, \ldots, g_m)$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Consider the exact sequence of complexes

$$0 \to I \otimes_A K_\bullet(A, g_1, \ldots, g_m) \to K_\bullet(A, g_1, \ldots, g_m) \to K_\bullet(A/I, g_1, \ldots, g_m) \to 0$$

Since the complex on the right has $H_1 = 0$ by assumption we see that

$$\mathrm{Coker}(I^{\oplus m} \to I) \longrightarrow \mathrm{Coker}(A^{\oplus m} \to A)$$

is injective. This is equivalent to the assertion of the lemma. □